ScholarPi: A Transparent, Deterministic-First Framework for Research Assessment

Design, rubric, and honest limitations of a CoARA-aligned manuscript assessment system

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ABSTRACT

Research assessment increasingly relies on quantitative indicators that are neither published nor auditable, and that measure the venue a paper appeared in rather than what the paper reports. ScholarPi is an open assessment framework that inverts this: it scores manuscripts against eight CoARA-aligned criteria in which 78% of the composite derives from deterministic, reproducible text analysis and only 22% from language-model interpretation, and it publishes the full rubric, every weight, and the split between the two. Large language models are used as a panel of jurors rather than as an oracle; their contribution is bounded by weight, gated on genuine cross-provider corroboration, and explicitly labelled where it cannot be verified. Two of the eight criteria are marked interpretive because novelty and societal impact are relations between a manuscript and its field or its future, not properties recoverable from a PDF. Every assessment is written to an append-only Proof-of-Research chain, and the scoring rubric is versioned so historical scores stay interpretable. This paper describes the architecture, states the rubric in full, and reports the system's limitations without mitigation: at the time of writing the reference deployment had assessed a single manuscript, no validation study against expert judgement has been conducted, and the framework therefore has no demonstrated accuracy. What it offers instead is auditability — every number it produces can be traced to a named, published signal.

1. Introduction

Quantitative research assessment has a well-documented failure mode. The San Francisco Declaration on Research Assessment [1] identified it precisely in 2013: the Journal Impact Factor, designed as a library acquisition tool for comparing journals, became a proxy for the quality of individual articles and individual researchers. The Leiden Manifesto [2] followed in 2015 with ten principles for the responsible use of metrics, the first of which is that quantitative evaluation should support, not supplant, qualitative expert assessment. The Agreement on Reforming Research Assessment [3], finalised in July 2022 and now the basis of the CoARA coalition, commits signatories to recognising a diverse range of outputs and to abandoning inappropriate uses of journal- and publication-based metrics.

These documents agree on the diagnosis. What none of them supplies is an instrument. A reviewer who wants to check whether a manuscript reports its randomisation procedure, registers its reagents, deposits its data under an open licence, and cites literature that actually resolves, must do so by hand — a tedious, unrewarded task that is consequently done inconsistently or not at all.

ScholarPi is an attempt at that instrument, built on a deliberately narrow claim. It does not measure research quality. It measures reporting quality and research integrity: what a manuscript documents, registers, deposits, and cites. Those are properties of the document, they are verifiable, and checking them is exactly the kind of work a machine should do so that human reviewers can spend their attention on the questions that require judgement.

2. Design principles

Four commitments constrain every part of the system, and several of them cost capability. They are stated here because the trade-offs they impose explain design decisions that would otherwise look arbitrary.

- **Deterministic first.** Where a property can be measured from the text, it is measured, not inferred. Model opinion is a fallback for what cannot be counted, not a default. The rubric quantifies this: 78% of the composite is deterministic.
- **Published methodology.** Every criterion is a weighted sum of named signals, every weight is stated, and the rubric is served from a versioned API endpoint. CoARA's transparency commitment is not satisfiable when the methodology is an undocumented coefficient buried in a function body.
- **Bounded claims.** Where the system cannot measure something, it says so rather than producing a number to two decimal places. Two criteria are flagged interpretive; corroboration is reported as a tier, not a score; a single-juror verdict is labelled as uncorroborated.
- **Auditability over accuracy.** Given a choice between a more accurate score nobody can check and a less accurate score anyone can reconstruct, the framework takes the second. A metric that cannot be audited cannot be contested, and a metric that cannot be contested cannot be corrected.

3. System architecture

3.1 Intake and extraction

Manuscripts enter as uploaded PDFs, by DOI resolution through Unpaywall, Semantic Scholar and CORE, or by discovery through OpenAlex [4]. Text is extracted with layout awareness: the largest-and-highest text block is not reliably the title, and the line below it is frequently a journal banner rather than an author list, so candidate blocks are scored on several weak signals (font size relative to body text, boldness, vertical position) rather than taken positionally.

Bibliographic metadata is reconciled across three tiers in order of authority: publisher-deposited registry metadata from Crossref or OpenAlex, then PDF typography, then the model panel. Author bylines are terminated at the first line that is a place or a date, because a title block is conventionally Title / Authors / Affiliation / City, Country / Date and the affiliation markers alone do not match a bare “Milan, Italy”.

3.2 Deterministic signals

Thirteen signals are measured, each normalised to [0, 1] with a stated meaning. The four that carry the most weight are MDAR reporting adherence, following the Materials Design Analysis Reporting framework [5]; density of valid Research Resource Identifiers [6], saturating at five; open-science reproducibility artefacts (repository, data availability statement, open licence, container, preregistration); and empirical density, the concentration of statistical reporting, sample sizes and quantitative results.

Reference integrity is measured by resolving cited DOIs against OpenAlex and Crossref. This matters more than it may appear: a fabricated citation is the highest-cost extraction error the system can make, and resolvability is a cheap, decisive check on it.

3.3 The juror panel and corroboration

Five language-model jurors of different lineages assess each manuscript independently, alongside a deterministic structural analyser. Each juror has an ordered chain of candidate routes across independent providers, so a single account-level restriction degrades a juror to its next route rather than removing it from the panel.

Corroboration is measured by the number of distinct provider-and-model routes that contributed, not by the number of juror labels that answered. This distinction is load-bearing. Every juror chain terminates in a shared fallback model, so a constrained deployment can have five jurors all served by one model; counting that as five independent opinions would report strong corroboration for what is one model voting five times. Correlated agreement is not evidence, and the system reports the collapse explicitly when it occurs.

A per-evaluation canary token is issued and checked on return. A juror that emits the canary has been successfully prompt-injected by the manuscript, and logic integrity for that assessment is set to zero.

4. The Pi-Index rubric

Eight criteria, each a weighted sum of normalised signals whose weights sum to exactly 1.0. The composite is therefore bounded to [0, 100] by construction rather than by clamping — an earlier design used additive bonuses that could push a score past 100 and were then clipped, which silently discarded information.

The “deterministic share” column is the fraction of each criterion decided by verifiable text analysis rather than model opinion. It is computed from the rubric itself, so it cannot drift from the weights actually in force.

ID	Criterion	Principal signals	Det. share
C1	Semantic Originality *	panel 0.45, corroboration 0.25, citation engagement 0.20	0.30
C2	Methodological Rigor	MDAR 0.55, RRID density 0.20, ref. integrity 0.15	1.00
C3	Interdisciplinary Synergy	topic diversity 0.60, domain span 0.25	0.85
C4	Societal Reach *	open licence 0.30, topic diversity 0.25, domain span 0.20	0.75
C5	Open Science	reproducibility 0.60, licence 0.20, persistent IDs 0.20	1.00
C6	Literature Integration	citation engagement 0.40, ref. integrity 0.30	0.70
C7	Empirical Density	empirical density 0.70, MDAR 0.15	0.85
C8	Future Actionability	reproducibility 0.35, persistent IDs 0.30	0.75

** Interpretive. Aggregate: 77.5% verifiable, 22.5% model interpretation.*

C1 and C4 are marked interpretive and carry materially reduced deterministic weight. Novelty is a relation between a manuscript and its field; impact is a relation between a manuscript and the future. Neither is recoverable from the PDF. They are retained because they carry information a reader wants, but the composite reports them as informed opinion rather than measurement, and a confidence endpoint states the split numerically for every score produced.

C8 is grounded in the FAIR Guiding Principles [7]: findability, accessibility, interoperability and reusability are assessed through persistent identifiers, licensing, and the presence of reproducible artefacts.

5. Scilem: learned calibration under constraint

The four structural signals are combined by a five-parameter linear model — four weights and a bias — fitted online. The signals themselves are never learned: their value to the framework is precisely that they are reproducible and identical for identical input. Only their relative weighting adapts.

The model is deliberately small. The training set grows by one example per assessment, and anything with more capacity would memorise the corpus rather than learn from it. A five-parameter linear model over four bounded inputs can be printed in full, checked by hand, and explained to a reviewer, which matters more here than accuracy.

Two teachers supply the target, and they are not weighted equally. Panel consensus updates automatically but is refused below two genuinely distinct routes, for the reason given in §3.3: learning from an uncorroborated verdict teaches imitation of whichever model happened to answer, not assessment of research. Explicit human corrections carry a higher learning rate but are restricted to signed-in identities and bounded per paper, because a repeatable anonymous correction endpoint is a direct route to steering the scoring model. Every update is bounded, weights are held non-negative and renormalised, and the full observation log is retained so any state can be recomputed and audited from scratch.

6. Proof-of-Research ledger and piQ emission

Each assessment appends a block recording the evaluation hash, the criteria weighting that assessment implied, the rubric fingerprint, and the predecessor's hash. The chain is append-only: withdrawing a paper removes it from the corpus and all listings but leaves its block standing, because deleting a block would invalidate every successor and destroy the integrity claim the chain exists to make.

The genesis block is derived from the deployment's own identity — owner wallet, token contract, and chain ID — rather than being a fixed constant. With a hardcoded genesis, every instance would share an identical chain root and two exports from two different deployments would be cryptographically indistinguishable.

piQ, the contribution token, is emitted under a published difficulty schedule with three multiplicative mechanisms: a halving every 2,500 assessed papers to a floor of 1/16; a minimum piX threshold rising from 40 toward an asymptote of 62 as the corpus grows; and per-author diminishing returns to prevent volume farming. The full schedule is exposed through an endpoint so a researcher can compute in advance exactly what a given paper would earn.

7. Epoch weighting and forecasting

Each assessed manuscript's evidence profile implies a criteria weighting, recorded per block. The resulting series is projected one step ahead to show which criteria the assessed corpus is producing the strongest and most consistent evidence for.

The default projection is Holt's linear trend method [8], implemented in NumPy. This is a deliberate choice over the neural alternative: Holt's method models level and trend with two parameters, which is the correct complexity for a series of a few dozen points, and fitting anything heavier to that much data estimates noise. An LSTM path exists behind a configuration flag and a wall-clock budget, and degrades to the statistical projection on any failure or overrun. On series this short the two largely agree.

8. Threat model

The system assumes an adversarial submitter. Manuscripts are structurally validated before any inference is purchased; hidden text and metadata manipulation are scanned for; a per-evaluation canary detects prompt injection; anonymous submissions require proof of work; and free allowance is metered per distinct document fingerprint so resubmitting the same paper costs nothing and cannot drain another user's quota.

Provider errors are classified into a small set of neutral categories before reaching a user. A raw error naming the vendor, disclosing that a routing account exists, and linking to its settings page is not a researcher's concern and is useful to someone probing the deployment.

9. Limitations

This section is deliberately unmitigated. The framework's central claim is auditability, and a limitations section that argues its own limitations away would contradict it.

- **No validation study.** The framework has never been evaluated against expert human judgement. No inter-rater agreement, no correlation with peer-review outcomes, no predictive validity has been established. Its scores should be read as structured descriptions of reporting practice, not as assessments whose accuracy is known.
- **Corpus of one.** At the time of writing, the reference deployment had assessed a single manuscript (n = 1). Every corpus-relative mechanism — the rising quality bar, epoch weighting, field comparisons in the research assistant, the map of science — is therefore operating on essentially no data and should not be read as informative.
- **Reporting quality is not research quality.** A methodologically weak study that reports itself impeccably will score well. A brilliant one published without a data statement will not. This is a known and accepted property of what the rubric measures, not a defect to be tuned away, but it is fatal if the score is read as a quality judgement.
- **Language and discipline bias.** MDAR originates in the life sciences [5], and RRIDs are principally a biomedical convention [6]. Mathematics, theoretical physics and the humanities will score structurally lower on C2 and C5 for reasons that have nothing to do with their rigour. No discipline-relative normalisation is currently applied.
- **Model opinion remains 22%.** Reduced, bounded, and labelled — but not eliminated. Jurors share overlapping training corpora, so their agreement rules out idiosyncratic error but not systematic error common to all of them.
- **Extraction is imperfect.** Scores depend on text extracted from PDFs of varying quality. A scanned or unusually typeset manuscript will yield weaker signals, and the text-completeness signal flags but does not repair this.

10. Related work

ScholarPi's rubric operationalises commitments articulated in DORA [1], the Leiden Manifesto [2] and CoARA [3], none of which prescribe an instrument. Its deterministic checks build directly on MDAR [5], RRID [6] and FAIR [7]. Automated reporting-checklist tools exist in the life sciences and share the goal of relieving reviewers of mechanical verification; ScholarPi differs in publishing its full weighting, in quantifying the share of each score that rests on model opinion, and in binding each assessment to an append-only ledger with a versioned rubric fingerprint, so a score computed today remains interpretable after the rubric changes.

11. Conclusion

ScholarPi is a working instrument for a problem that has been well described and poorly tooled. Its contribution is not accuracy, which is unmeasured, but structure: an assessment in which every number traces to a named signal, the interpretive portion is quantified rather than hidden, and the whole rubric is published and versioned.

The most useful next step is the one this paper cannot substitute for: a validation study against expert judgement on a corpus large enough to support conclusions. Until that exists, the honest description of this framework is a transparent, auditable, and unvalidated one.

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